4 (a)(i)	The electric force per *unit* ***positive*** charge placed at that point.	[2]
(a)(ii)	Electric field strength at a point is equal to the negative potential gradient at that pt {Accept use of eqn, E = - $\frac{dV}{dx}$, as symbols are already defined in qn; cannot use '*r*' in place of '*x*'}	[1]
(a)(iii)	From the graph, at *x* = 15.0 cm, the gradient is positive. The electric field strength is negative, which means that it is pointing leftwards.	[1]
(b)	At *x* = 15.0 cm, the potential due to both spheres is zero. $V_A + V_B = 0$ $\frac{Q_A}{4\pi\varepsilon_o x} + \frac{Q_B}{4\pi\varepsilon_o (d-x)} = 0$ $\frac{Q_A}{0.15} = \frac{-0.48\times10^{-9}}{0.60-0.15}$ $Q_A = -1.6\times10^{-10}$ C	[1] [1] [1]
(c)	loss in electric potential energy = gain in kinetic energy $U_i - U_f = K.E._f - K.E._i$ $-1.60\times10^{-19}\times(0-140) = \frac{1}{2}\times9.11\times10^{-31}\times\left[v_f^{\ 2} - \left(4.0\times10^6\right)^2\right]$ $v_f \approx 8.1\times10^6$ m s^{-1}	[1] [1] [1]